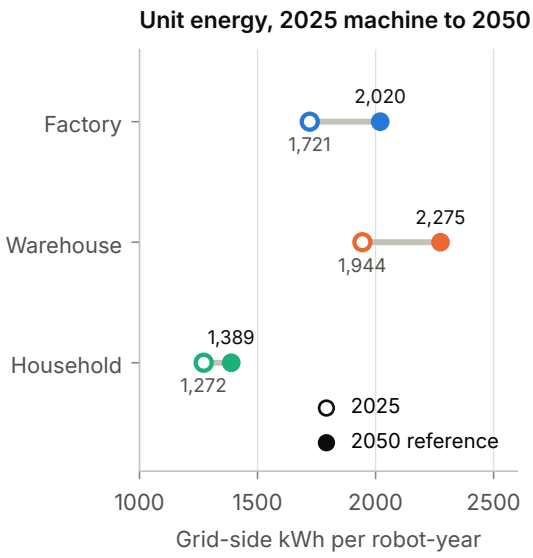
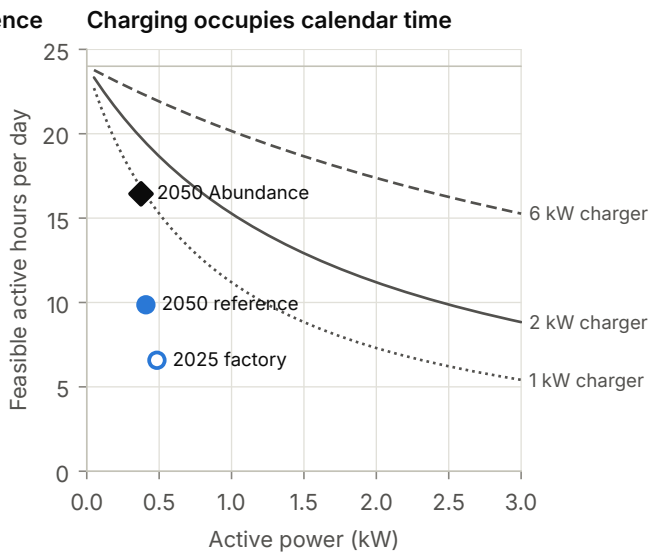

Unit energy, 2025 machine to 2050 reference
Factory
Warehouse
Household
2,020
1,721
2,275
1,944
1,389
1,272
2025
2050 reference
1000
1500
2000
2500
Grid-side kWh per robot-year


Charging occupies calendar time
Feasible active hours per day
25
20
15
10
5
0
2050 Abundance
2050 reference
2025 factory
6 kW charger
2 kW charger
1 kW charger
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Active power (kW)